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OPTICS.

tre is C, and focus of parallel rays is at P; and let the imaginary focus of the incident rays be as F, a point between P and B, and he DR be the refleth ray. From C and R draw the lines CH, RH, the one palling through D, the other parallel to FE. Then will the angle RHD be equal to HDD the angle of incidence, as alterant to is; and therefore equal to HDD, the angle of reflections is reflected the triangle HDR is indicates, and consequently DR is equal to RH. Now the lines FD and RH being parallel, the complex DDC and RHD are similar; and therefore RH/ or RD, is to PD as CR on CF: but BD vanthing, RD and FD coincide with RB and IB, whether RF is to TR as CR in CF; that is, the distance of the focus from the surface is to the dilance of the imaginary focus of the incident rays, at the distance of the focus from the centre is to the dilance of the imaginary focus of the incident rays from the fame. _@E.D._

or Refletions.

Cass. Of diverging rays falling upon a concave surface, and corresponding from a point between the fo-cus of parallel rays and the surface.

Dass. Let ED and FB represent two diverging rays flowing from the point 3 as a radiant, which was the imaginary focus of the incident rays in the forgoing cafe. Then an SD as in the cafe selected with anDR, FD will be reflected into DR(, (for the region unem- tioned in Cafc the feecond), lo that the refleted ray will proceed as from the point 3, but it cane demon- flatred in the cafe manufacturing forceing, that RB is to FB as CR to CF; that is, the distance of the fo-cus from the surface is to that of the radiant from the fame, as the distance of the former from the centre is to that of the latter from the fame. _@E.D._

Cass 7. Of converging rays falling upon a convex surface, and including towards a point beyond the centro.

Dass. Let AB, ED, (fig. 8) be the incident rays tending to F, a point beyond the centre C, and let DR be the refleted ray of the incident one ED. Then because the incident ray ED tend to a point beyond the centre, the refleted ray DR will proceed as from one on the contrary to, flopode 2 as as- related above under Drop, VII. Through D draw the perpendicular CD, and produce it to H. Thus will EDH and HDD be the angles of incidence and re- flelections, while being equal, their vertical ones of DF and CDR will be fo to cases confiantly the vertex of the triangle FDR and bifled by the line CD+ where- fore, RD is to DF, or (3 Elem d o) BD vanthing, RB is to DF as RC to CF; that is, the dilatance of the focus of the refleted rays is to that of the imaginary focus of the incident rays, as the dilatance the lower from the center is to the dilatance of the latter from the fame. _@E.D._

Cass 8. Of diverging rays falling upon a concave surface, and proceeding from a point beyond the centro.

Dass. Let FB, ED, be the incident rays having their radiant in F, the imaginary focus of the incident rays in the foreground cafe. Then an ED was in that cafe reflected into DR(, FD will now be reflected into

DR; lo that R will be the focus of the refleted rays. But it was demontrated in the foregoing cafe, that RB is to FB as RC to CF; that is, the dilatance of

Part II.

the focus of the refleted rays from the furface is to the dilatance of the radiant from the fame, as the di- flance of the focus of the refleted rays from the cen- tre is to the dilatance of the radiant from thence. _@E.D._

The two remaining cafes may be confirmed as the converfe of thote under Prop. II. (p. 548-), because the incident rays in there are the reflected ones in them; or they may be demothered in the fame manner with the forgoing, as follows.

Cass 9. Converging rays falling upon a convex fur- face, and tending to the surface of parallel rays, become parallel after reflection.

Dass. Let ED, RB, (fig. 8) reported two convergency rays inside to the convex fur- End, and tending towards F, which we will now loppole to be the focus of parallel rays; and he DR be the refleted ray and Cash center of coronavity of the re- flefling surfaces. Through C draw the line CD, and produces to FR, drawing ARH parallel to ED prod-cut to F. Now it has been demottemical Cade ? when the incident rays are loppodof to tend to the point Y, that RB is to FB as RC to CF; but F in the Cafe being loppold to be the focus of parallel rays, it is the middle point between C and B (by pro- position all), and therefore FB and FC are equal and confiantly the two outer corners in the proportion, viz. RB and RC, mitb fe toto ; which can only be upon a loppodiation that R is at an infinite diffuse from B; that is, that the refleted ray DR and DR be parallel. _@E.D._

Cass to Diverging rays falling upon a concave fur-ness and preceding from the focus of parallel rays, become parallel after reflection.

Dass. Let AD, EB, BB (fig. 6) be two diverging rays incident upon the concave fur BD, as loppo- fed in Cafe 4; where it was demottemical that FB is to RB as CF to OR, then in the prefect each RB and CR are equal, because R is loppold to be the focus of parallel rays; therefore FB and FC are fo too. Which cannot be under F be taken as the nature di- fame from B; that is, and the refleted rays BF and DF be parallel. _@E.D._

It is here otherable, that in the cafe of diverging rays falling upon a convex furace, (fe fig. 7) the further the point It is taken from B, not never the point F, the focus of the refleted rays, approaches to B, while the radiant R remains the fame. For is in evident from the curvature of events, thus the ripet D (fig. 9) may be te ken fo far from B, that the refleted ray DE final proceed as from F, G, H, or even from B, or from points between B and R; and the further is in taken from B, the father the point from which it pro- cocis, appochoz toward R is will easily appear if we doue forced incident rays with that respec- tive reflected ones, in fashion than that angles of reflection may be all equal to their refleting angles of incidentes as at those in the figure. The fact is applicable to any of the other cafes of diverging or converging rays incident upon a spherical surface. That is the reason, that when rays are considered as es- flected from a spherical surface, the dilatance of the oblique rays from the perpendicular one is taken to [all, that it may be loppold to vanish.

From

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Of Reflection.

tre is C, and focus of parallel rays is at P; and let the imaginary focus of the incident rays be at F, a point between P and B; and let DR be the reflected ray. From C and R draw the lines CH, RH, the one paſſing through D, the other parallel to FE. Then will the angle RHD be equal to HDE the angle of incidence, as alternate to it; and therefore equal to HDR, the angle of reflection: wherefore the triangle HDR is iſoſceles, and conſequently DR is equal to RH. Now the lines FD and RH being parallel, the triangles FDC and RHC are ſimilar; and therefore RH, or RD, is to FD as CR to CF: but BD vaniſhing, RD and FD coincide with RB and FB, wherefore RB is to FB as CR to CF; that is, the diſtance of the focus from the ſurface is to the diſtance of the imaginary focus of the incident rays, as the diſtance of the focus from the centre is to the diſtance of the imaginary focus of the incident rays from the ſame. *Q. E. D.*

Case 6. Of diverging rays falling upon a concave ſurface, and proceeding from a point between the focus of parallel rays and the ſurface.

Dem. Let FD and FB repreſent two diverging rays flowing from the point F as a radiant, which was the imaginary focus of the incident rays in the foregoing caſe. Then as ED was in that caſe reflected into DR, FD will be reflected into DK, (for the reaſon mentioned in Caſe the ſecond), ſo that the reflected ray will proceed as from the point R: but it was demonſtrated in the caſe immediately foregoing, that RB is to FB as CR to CF; that is, the diſtance of the focus from the ſurface is to that of the radiant from the ſame, as the diſtance of the former from the centre is to that of the latter from the ſame. *Q. E. D.*

Case 7. Of converging rays falling upon a convex ſurface, and tending towards a point beyond the centre.

Plate CCXII.

Dem. Let AB, ED, (fig. 8.) be the incident rays tending to F, a point beyond the centre C, and let DK be the reflected ray of the incident one ED. Then becauſe the incident ray ED tends to a point beyond the centre, the reflected ray DK will proceed as from one on the contrary ſide, ſuppoſe R; as explained above under Prop. VII. Through D draw the perpendicular CD, and produce it to H. Then will EDH and HDK be the angles of incidence and reflection; which being equal, their vertical ones CDF and CDR will be ſo too: conſequently the vertex of the triangle FDR is biſected by the line CD: wherefore, RD is to DF, or (3 Elem. 6.) BD vaniſhing, RB is to BF as RC to CF; that is, the diſtance of the focus of the reflected rays is to that of the imaginary focus of the incident rays, as the diſtance of the former from the centre is to the diſtance of the latter from the ſame. *Q. E. D.*

Case 8. Of diverging rays falling upon a concave ſurface, and proceeding from a point beyond the centre.

Dem. Let FB, FD, be the incident rays having their radiant in F, the imaginary focus of the incident rays in the foregoing caſe. Then as ED was in that caſe reflected into DK, FD will now be reflected into DR; ſo that R will be the focus of the reflected rays. But it was demonſtrated in the foregoing caſe, that RB is to FB as RC to CF; that is, the diſtance of the focus of the reflected rays from the ſurface is to the diſtance of the radiant from the ſame, as the diſtance of the focus of the reflected rays from the centre is to the diſtance of the radiant from thence. *Q. E. D.* Of Reflection.

The two remaining caſes may be conſidered as the converſe of thoſe under Prop. II. (p. 5548.), becauſe the incident rays in theſe are the reflected ones in them; or they may be demonſtrated in the ſame manner with the foregoing, as follows.

Case 9. Converging rays falling upon a convex ſurface, and tending to the ſurface of parallel rays, become parallel after reflection.

Dem. Let ED, RB, (fig. 7.) repreſent two converging rays incident on the convex ſurface BD, and tending towards F, which we will now ſuppoſe to be the focus of parallel rays; and let DR be the reflected ray, and C the centre of convexity of the reflecting ſurface. Through C draw the line CD, and produce it to H, drawing RH parallel to ED produced to F. Now it has been demonſtrated (Caſe 5.) where the incident rays are ſuppoſed to tend to the point F, that RB is to FB as RC to CF; but F in this Caſe being ſuppoſed to be the focus of parallel rays, it is the middle point between C and B (by propoſition 2d), and therefore FB and FC are equal; and conſequently the two other terms in the proportion, viz. RB and RC, muſt be ſo too; which can only be upon a ſuppoſition that R is at an infinite diſtance from B; that is, that the reflected rays BR and DR be parallel. *Q. E. D.*

Case 10. Diverging rays falling upon a concave ſurface, and proceeding from the focus of parallel rays, become parallel after reflection.

Dem. Let RD, RB (fig. 8.) be two diverging rays incident upon the concave ſurface BD, as ſuppoſed in Caſe 4.; where it was demonſtrated that FB is to RB as CF to CR. But in the preſent caſe RB and CR are equal, becauſe R is ſuppoſed to be the focus of parallel rays; therefore FB and FC are ſo too. Which cannot be unleſs F be taken at an infinite diſtance from B; that is, unleſs the reflected rays BF and DF be parallel. *Q. E. D.*

Obs. It is here obſervable, that in the caſe of diverging rays falling upon a convex ſurface, (ſee fig. 7.) the farther the point D is taken from B, the nearer the point F, the focus of the reflected rays, approaches to B, while the radiant R remains the ſame. For it is evident from the curvature of a circle, that the point D (fig. 9.) may be taken ſo far from B, that the reflected ray DE ſhall proceed as from F, G, H, or even from B, or from any point between B and R; and the farther it is taken from B, the faſter the point from which it proceeds, approaches towards R: as will eaſily appear if we draw ſeveral incident rays with their reſpective reflected ones, in ſuch manner that the angles of reflection may be all equal to their reſpective angles of incidence, as is done in the figure. The like is applicable to any of the other caſes of diverging or converging rays incident upon a ſpherical ſurface. This is the reaſon, that when rays are conſidered as reflected from a ſpherical ſurface, the diſtance of the oblique rays from the perpendicular one is taken ſo ſmall, that it may be ſuppoſed to vaniſh.

From

Figure 1: *
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